

Direct Retainers

Learning objectives:

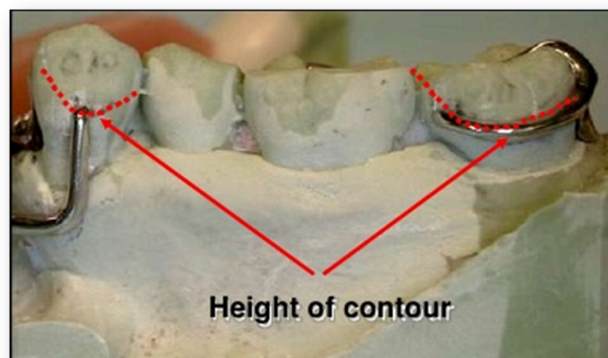
At the end of this lecture, the student should be able to

- Distinguish between different types of direct retainers
- Discover the main requirements of direct retainers
- Identify the advantages, disadvantages, and indications of each type.
- Recognise the factors affecting the magnitude of retention in direct retainers.
- Apprehend the most common types of direct retainers used for tooth-borne RPD.

A direct retainer is any unit of a removable dental prosthesis that engages an abutment tooth to prevent the prosthesis from being displaced away from the basal seat tissue.

There are two classes of direct retainers: extracoronal and intracoronal.

- **Intracoronal retainers (precision attachments)**
- **Extra-coronal direct retainers** engage the external surface of an abutment in a natural undercut or a prepared depression. There are two main classes of clasps:
 - 1) those that approach the undercut from above the height of the contour (supra bulge retainers)and 2) those that approach the undercut from below (infra bulge retainers).



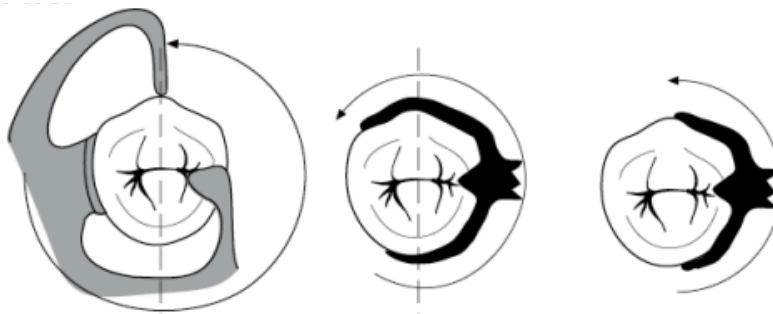
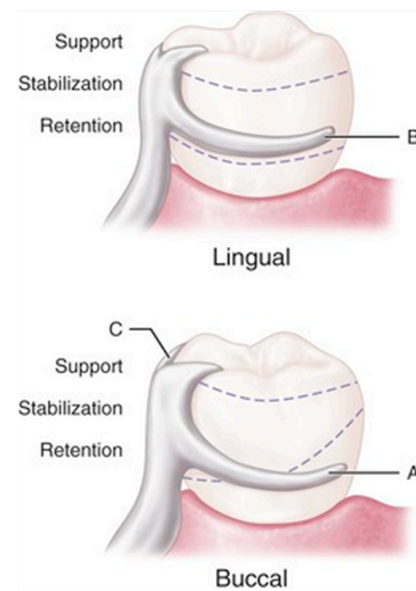
Extra-Coronal Direct Retainers

It usually comprises a retentive arm, a reciprocal (bracing) element or arm, a rest and a minor connector. Retention is derived by placing a clasp arm into an undercut area so that it is forced to deform upon vertical dislodgment. The resistance of the clasp to deformation generates retention. Resistance is proportionate to the flexibility of the clasp arm. Non-flexible portions of clasp arms must be placed occlusal to the height of the contour (supra-bulge area).

Requirements of Extra-coronal Direct Retainers

All clasp assemblies should meet the following requirements:

1. **Support** - resistance to gingival displacement (occlusal rests)
2. **Reciprocity** - resistance to orthodontic movement of teeth using reciprocal arms. During placement and removal of the partial denture, the retentive arm flexes over the height of the contour and generates energy. At this point, the rigid reciprocal arm should contact the guiding plane and prevent orthodontic movement.
3. **Stability** - resistance to lateral movement (reciprocal arms, minor connectors)
4. **Retention** - retentive arms located in undercuts on the abutments
5. **Encirclement of more than 180° of the tooth** prevents the prosthesis from moving away from it.
6. **Passivity** - at rest, a direct retainer should not exert force against a tooth



Factors affecting the magnitude of retention

1. Size of the angle of convergence.



2. How far into the angle of convergence the clasp terminal is placed?

When the angle of convergence between two abutments differs, uniformity of retention can be obtained by placing the clasp arms into the same degree of undercut (i.e. both 0.01"). A guiding principle of partial denture design is that retention should be uniform in magnitude and bilaterally opposed among abutments.



3. Flexibility of the clasp arm. This is influenced by the following factors:

- i. Length: increased length increases flexibility
- ii. Diameter: diameter is inversely proportional to flexibility
- iii. Cross-sectional form
 - a. round forms are usually more flexible (wrought or cast)
 - b. Half-round shape is limited to flexure in only two directions (cast)



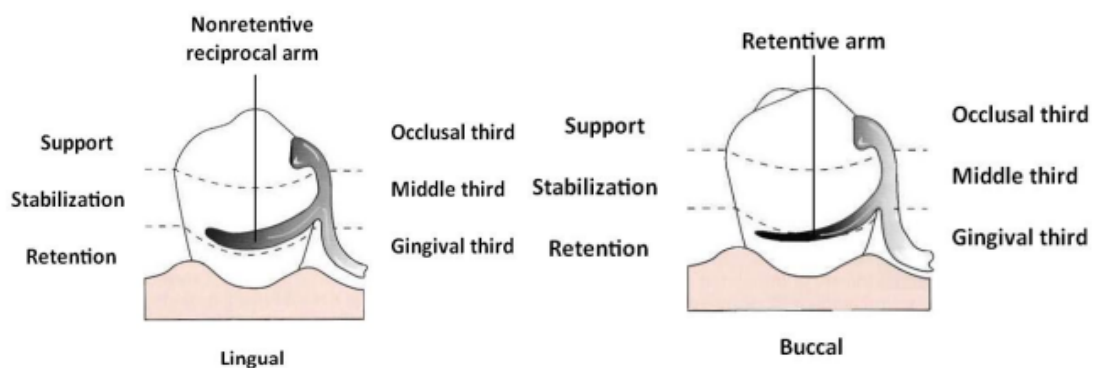
iv. Clasp material: Wrought wire clasps have greater tensile strength than cast clasps and hence can be used in smaller diameters to provide greater flexibility without fatigue or fracture.

Direct Retainers For Tooth-Borne Removable Partial Dentures

Clasps for tooth-borne partial dentures (Class III, IV) have one function – to prevent dislodgment of the prosthesis without damage to the abutment teeth. Since there is little or no rotation caused by tissue-ward movement of the edentulous area (as happens in distal extension cases), stress-releasing properties are usually not required. These clasps can also be used in modification spaces for tooth and tissue-supported removable partial dentures (Class I, II).

1. Circumferential (Circle or Akers) clasp

- a. the most simple and versatile clasp (clasp of choice in tooth-borne cases)
- b. clasp assembly has one retentive arm opposed by a reciprocal arm originating from the rest
- c. the retentive arm begins above the height of contour, and curves and tapers to its terminal tip, in the gingival 1/3 of the tooth, well away from the gingiva
- d. the bracing arm is in the middle 1/3 of the tooth, is broader occluso-gingivally, does not taper and is either entirely above the height of contour or completely on a prepared guiding plane – it should never be designed into an undercut, as it is a rigid element.



Advantages:

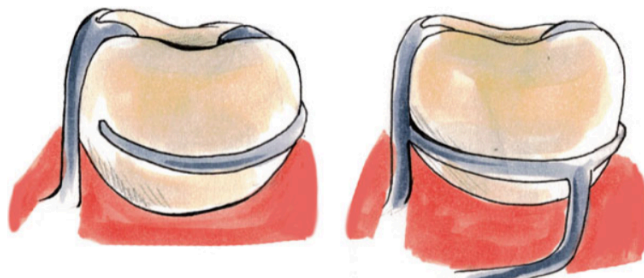
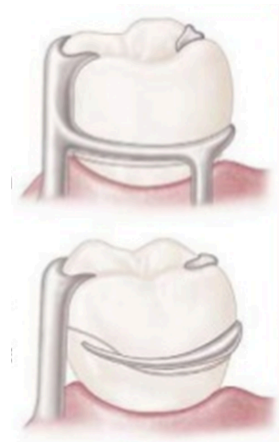
- a. Excellent bracing qualities.
- b. Easy to design and construct.
- c. Less potential for food accumulation below the clasp compared to bar clasps.

Disadvantages:

- a. More tooth coverage than bar clasps.
- b. More metal is displayed than with bar or combination clasps.
- c. Adjustments are difficult or impossible due to the half-round nature of the clasp.

2. Ring clasp

- a. Encircles nearly the entire abutment tooth
- b. The undercut is adjacent to the edentulous span.
- c. Should always be used with a supporting strut on the non-retentive side with an auxiliary occlusal rest on the opposite side. The omission of the supporting strut will allow the clasp arm to open and close with minimal or no reciprocation.

**Advantages:**

- a. Excellent bracing (with supporting strut).
- b. Allows use of an available undercut adjacent to the edentulous area.

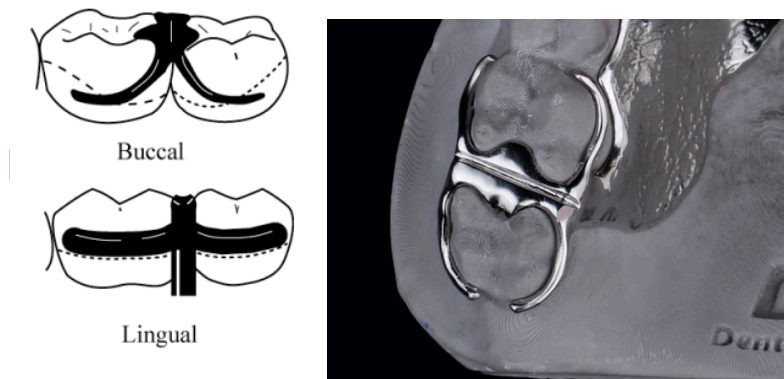
Disadvantages:

- a. Covers a large area of the tooth surface, therefore requiring meticulous hygiene.
- b. Very difficult to adjust due to the extreme rigidity of the reciprocal arms.

Contraindications: excessive tissue undercuts prevent the use of a supporting strut.

3. Embrasure (Double Akers) Clasp

- Two rests, two retentive arms, and two bracing arms
- Used in a quadrant where no edentulous area exists, or where a distal approach clasp cannot be used on the most posterior tooth (i.e. no usable retentive undercut).
- Use with discretion – use another clasp if possible.



Advantage:

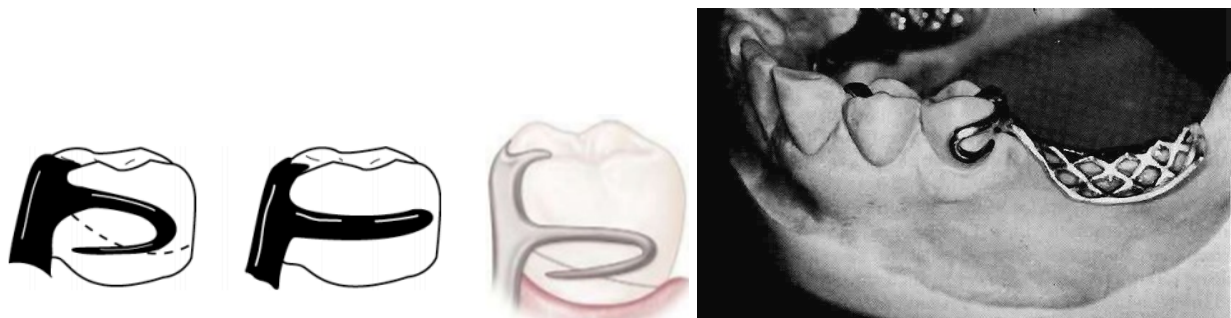
- Allows placement of a direct retainer where none could otherwise be placed (especially contralateral to the edentulous span on a Class II case)

Disadvantages:

- a. Extensive interproximal reduction is usually required
- b. Covers a large area of tooth surface - hygiene considerations

4. Hairpin or Reverse action

The retentive area (undercut) is adjacent to the occlusal rest.



Advantage:

- Allows use of undercut adjacent to the edentulous space

Disadvantages:

1. Non-esthetic
 2. Almost impossible to adjust
 3. Covers an extensive tooth surface and acts as a food trap
 4. Insufficient flexibility on short crowns due to insufficient clasp arm length
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5. Back action clasp

- a. The back-action clasp is a modification of the ring clasp.
- b. It is used on a premolar abutment anterior to the edentulous space.
- c. The undercut can usually be approached just as well using a conventional circumferential clasp, with less tooth coverage and less display of metal.

**6. Multiple clasp**

The multiple clasp is simply two opposing circumferential clasps joined at the terminal end of the two reciprocal arms.

Indications:

- a. It is used when additional retention and stabilisation are needed, usually on tooth-supported partial dentures.
- b. It may be used for multiple clasping in instances in which the partial denture replaces an entire half of the dental arch.
- c. It may be used rather than an embrasure clasp when the only available retentive areas are adjacent to each other.

**Disadvantage:**

- a. Its disadvantage is that two embrasure approaches are necessary rather than a single common embrasure for both clasps.
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7. Half-and-half Clasp

a. It consists of a circumferential retentive arm arising from one direction and a reciprocal arm arising from another.

b. The second arm must arise from a second minor connector, and this arm is used with or without an auxiliary occlusal rest.

c. Reciprocation arising from a second minor connector usually can be accomplished with a short bar or with an auxiliary occlusal rest, thereby avoiding so much tooth coverage.

d. There is little justification for the use of the half-and-half clasp in bilateral extension base partial dentures.

